

Dynamic Analysis of Malware

Malware - <https://github.com/Da2dalus/The-MALWARE-Repo/tree/master/Virus/xpajB.exe>

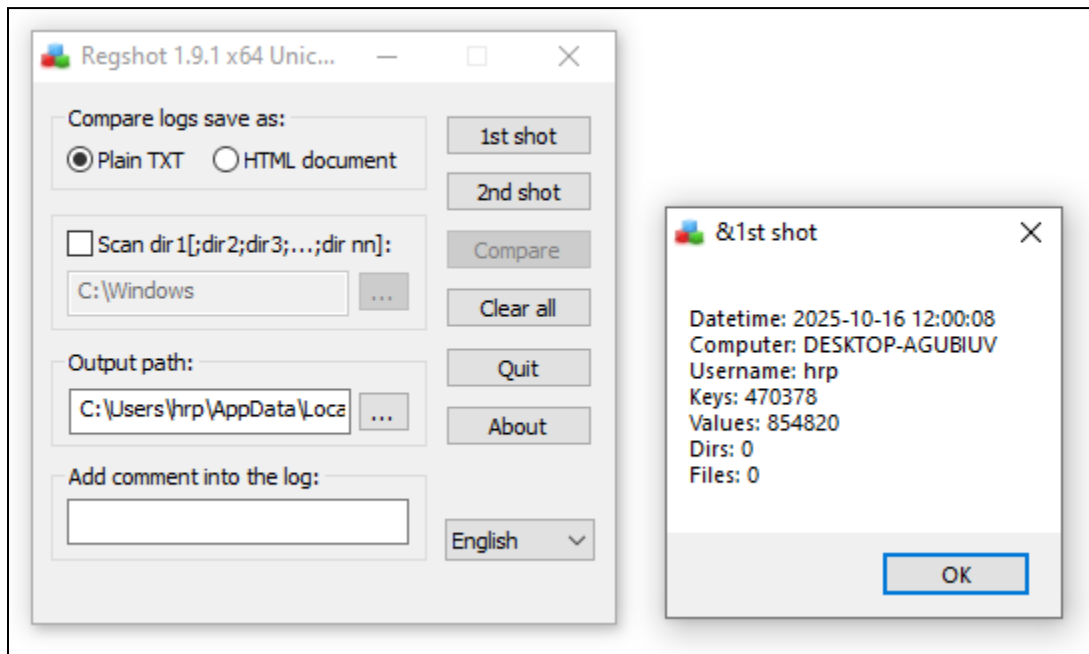
The targeted file is - xpajB.exe



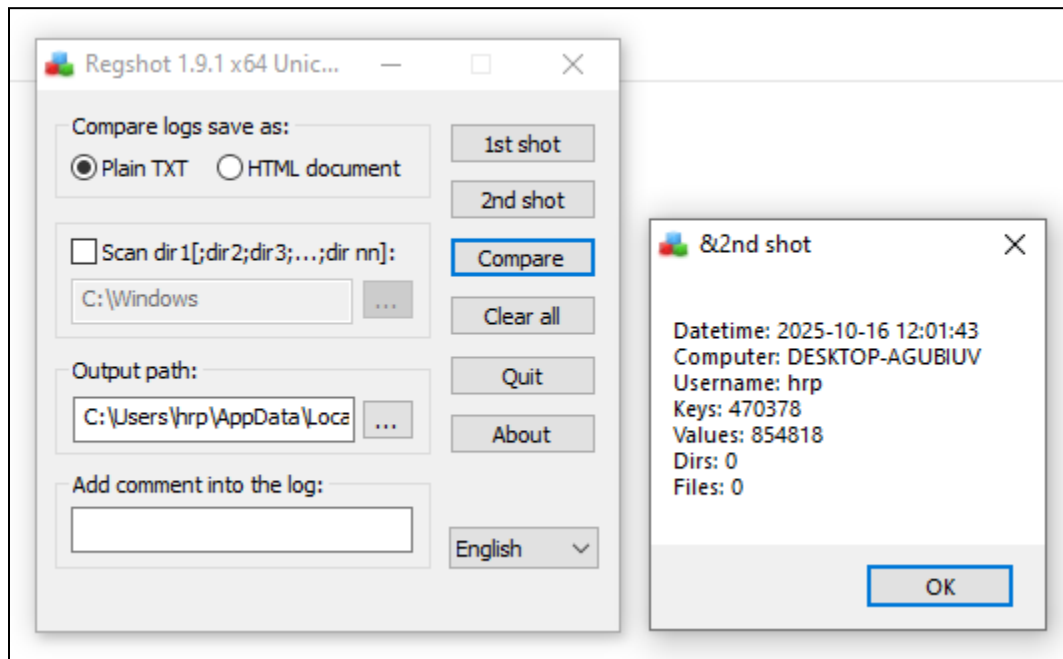
Initial Behavioral Observation

Tool used - Regshot

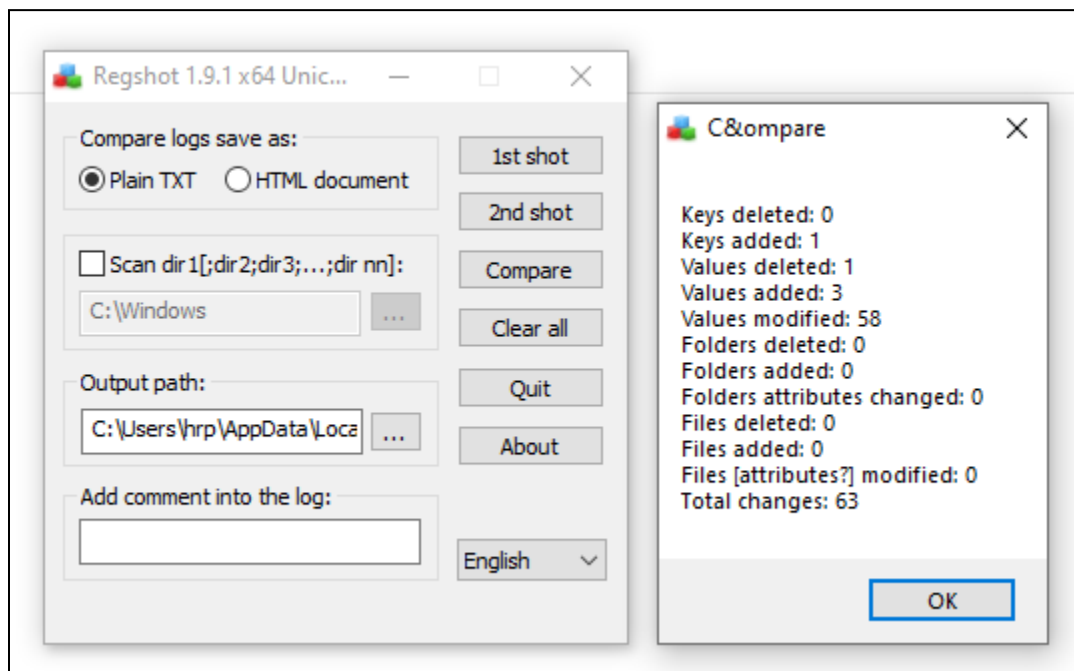
Before the execution of the malware:



After the execution of Malware:



Comparison of 1st shot & 2nd shot:



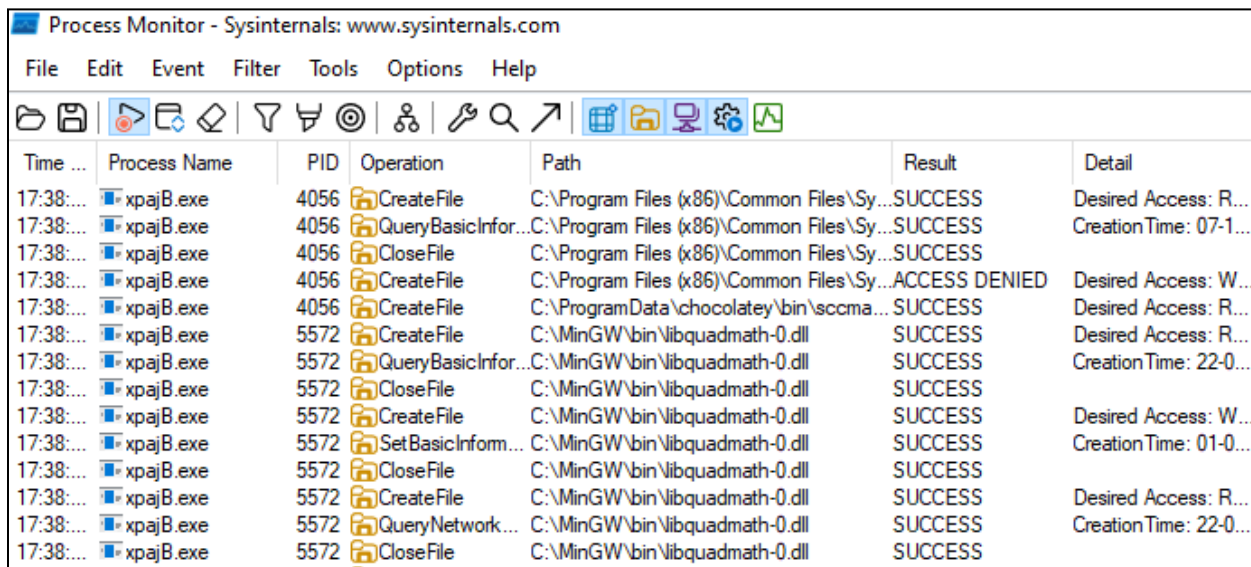
Observations:

- The malware was executed in a controlled virtual environment to monitor its initial behavior.
- The total number of registry keys remained constant at 470,378, showing no major new key creation.
- The total number of registry values changed slightly from 854,820 to 854,818, indicating minor modifications.
- No new files or directories were created during the malware execution period.
- The minimal registry and file system changes suggest that the malware made **limited system modifications**.
- The small variation in registry values indicates the possibility of **temporary or in-memory operations**.
- Overall, the malware appears to perform **lightweight or memory-based activity** without leaving strong traces on the system.

Process and Thread Monitoring

Tool used : Process Monitor

Filter : File name include xpajB.exe



The screenshot shows the Process Monitor application window with the filter 'File name include xpajB.exe'. The table below represents the data visible in the main pane.

Time ...	Process Name	PID	Operation	Path	Result	Detail
17:38:...	xpajB.exe	4056	CreateFile	C:\Program Files (x86)\Common Files\Sy...	SUCCESS	Desired Access: R...
17:38:...	xpajB.exe	4056	QueryBasicInfor...	C:\Program Files (x86)\Common Files\Sy...	SUCCESS	CreationTime: 07-1...
17:38:...	xpajB.exe	4056	CloseFile	C:\Program Files (x86)\Common Files\Sy...	SUCCESS	
17:38:...	xpajB.exe	4056	CreateFile	C:\Program Files (x86)\Common Files\Sy...	ACCESS DENIED	Desired Access: W...
17:38:...	xpajB.exe	4056	CreateFile	C:\ProgramData\chocolatey\bin\sccma...	SUCCESS	Desired Access: R...
17:38:...	xpajB.exe	5572	CreateFile	C:\MinGW\bin\libquadmath-0.dll	SUCCESS	Desired Access: R...
17:38:...	xpajB.exe	5572	QueryBasicInfor...	C:\MinGW\bin\libquadmath-0.dll	SUCCESS	CreationTime: 22-0...
17:38:...	xpajB.exe	5572	CloseFile	C:\MinGW\bin\libquadmath-0.dll	SUCCESS	
17:38:...	xpajB.exe	5572	CreateFile	C:\MinGW\bin\libquadmath-0.dll	SUCCESS	Desired Access: W...
17:38:...	xpajB.exe	5572	SetBasicInform...	C:\MinGW\bin\libquadmath-0.dll	SUCCESS	CreationTime: 01-0...
17:38:...	xpajB.exe	5572	CloseFile	C:\MinGW\bin\libquadmath-0.dll	SUCCESS	
17:38:...	xpajB.exe	5572	CreateFile	C:\MinGW\bin\libquadmath-0.dll	SUCCESS	Desired Access: R...
17:38:...	xpajB.exe	5572	QueryNetwork...	C:\MinGW\bin\libquadmath-0.dll	SUCCESS	CreationTime: 22-0...
17:38:...	xpajB.exe	5572	CloseFile	C:\MinGW\bin\libquadmath-0.dll	SUCCESS	

Observations:

Process Identification and Activity

- **Process Name:** The activity is dominated by the process **xpajB.exe**.
- **Process IDs (PID):** Two distinct instances of the process are shown operating concurrently:
 - **PID 4056:** Performing operations on files within a system path: C:\Program Files (x86)\Common Files\Sy...
 - **PID 5572:** Performing operations on files within a development/library path: C:\MinGW\bin\libquadmath-0.dll
 - **Monitoring Takeaway:** This indicates two separate instances (or perhaps a parent/child relationship) of the xpajB.exe program running, each accessing different areas of the file system.

File System Operations

While the requested task is "Process and Thread Monitoring," Procmon often shows many event types. The primary file system operations visible here are:

- **CreateFile:** An attempt to open or create a file. The **Detail** column shows the **Desired Access** (e.g., Read, Write).
- **QueryBasicInformationFile:** Getting metadata about a file, such as its creation and modification times.
- **CloseFile:** Closing an open file handle.
- **SetBasicInformationFile:** Attempting to set metadata for a file.
- **QueryNetworkOpenInformationFile:** Getting information potentially related to file sharing or network attributes of the file.

Key Findings

- **Access Denied:** An important event is visible for **PID 4056** with the **Operation** CreateFile and **Result** **ACCESS DENIED**. This means the process was blocked by Windows security permissions when attempting to create or open a file with specific access rights (likely write access). This is a crucial finding for troubleshooting or security analysis.
- **DLL Access:** **PID 5572** shows multiple operations (QueryBasicInformationFile, CloseFile, CreateFile) against the **libquadmath-0.dll** file in the MinGW directory. This confirms the process is **loading and interacting with this external library** at the time of the capture.

Events and Targeted Directories

The process **xpajB.exe (PID 7536)** is primarily performing **CreateFile** and **QueryBasicInformationFile** operations, which includes attempts to open, read, and check the existence of files.

Based on the provided Process Monitor screenshots (173856.png and 180218.png), here is the recording and analysis of file system activity by the process **xpajB.exe**, presented without tables.

1. Recorded File Creation, Modification, or Deletion Events

The execution of **xpajB.exe** is dominated by attempts to access and read files, with limited successful write operations observed in the visible logs:

- **File Creation/Access Attempts:** A vast number of **CreateFile** operations were recorded, primarily resulting in **NAME NOT FOUND** (e.g., PID 7536). This indicates the process is intensely **searching for files** that do not exist along the paths it is checking, primarily system DLLs.
- **Modification/Write Events (Blocked):** A critical event for **PID 4056** shows a **CreateFile** operation resulting in **ACCESS DENIED**. This signifies a **failed attempt to modify or create a file** in a protected directory (e.g., within **C:\Program Files (x86)\Common Files...**).
- **Successful Modification/Read Events:** The process successfully performs operations like **QueryBasicInformationFile** and **CloseFile** against various libraries, confirming **successful reading of file metadata and contents**.
- **Deletion Events:** No explicit file deletion events (**SetDispositionInformationFile**) were visible in the recorded segments of the log.

2. Specific Directories Targeted

The file activity shows clear targeting of critical system and user directories:

- **Core Windows System Directories:** The process performs massive, high-speed probing within **C:\Windows\System32** and **C:\Windows\SysWOW64**.
- **User Downloads Directory:** The path **C:\Users\hrp\Downloads** is explicitly targeted, with attempts to access specific files within it.
- **Application/Library Paths:** Other directories include **C:\MinGW\bin** and paths within **C:\ProgramData** (e.g., related to 'chocolatey').

3. Nature of Created/Accessed Files

The nature of the files involved points primarily to executable code and its dependencies:

- **System Dependencies / Probes:**
 - Nature: Predominantly System Libraries (.dll) (e.g., **winmm.dll**, **d3dcompiler_43.dll**).

- Role: The intensive search (resulting in NAME NOT FOUND) is characteristic of a program attempting to resolve dependencies or actively performing checks for DLL Search Order Hijacking.
- **Dropped Payloads / Auxiliary Components:**
 - Nature: Executables and libraries (.exe, .dll) found in the Downloads path (e.g., cabine.dll).
 - Role: These are strongly suspected to be secondary components or payloads that the main process loads to continue its malicious or main function execution.
- **Configuration/Persistence (Attempted):**
 - Nature: An attempted write operation in a restricted directory.

Registry and Persistence Mechanism

Tool used : Procmon

explorer.exe	3688	0.14		64.98 MB	DESKTOP-AGUBIUUV\hrp	Windows Explorer
SecurityHealthSystray.exe	6680			1.61 MB	DESKTOP-AGUBIUUV\hrp	Windows Security notification...
VBoxTray.exe	6944	0.01	56 B/s	2.52 MB	DESKTOP-AGUBIUUV\hrp	VirtualBox Guest Additions Tra...
ZoomIt64.exe	7012			1.75 MB	DESKTOP-AGUBIUUV\hrp	Sysinternals Screen Magnifier
xpajB.exe	4056	11.43		5.82 MB	DESKTOP-AGUBIUUV\hrp	Основная библиотека обще...
xpajB.exe	5572	2.27		3.87 MB	DESKTOP-AGUBIUUV\hrp	Основная библиотека обще...
ProcessHacker.exe	6928	0.46		17.05 MB	DESKTOP-AGUBIUUV\hrp	Process Hacker

xpajB.exe (5572) Properties

File

Основная библиотека общего доступа к приложениям
(UNVERIFIED) Корпорация Майкрософт
Version: 5.1.2600.5512
Image file name:
C:\Users\hrp\AppData\Local\Temp\1229.tmp

Process

Command line: "C:\Users\hrp\Downloads\xpajB.exe"

Current directory: C:\Users\hrp\Downloads\

Started: 4 minutes and 29 seconds ago (17:31:27 16-10-2025)

PEB address: 0x31c000 (32-bit: 0x31d000) Image type: 32-bit

Parent: explorer.exe (3688)

Mitigation policies: None

Protection: None

Permissions Terminate

Close

xpajB.exe (4056) Properties

File

Основная библиотека общего доступа к приложениям
(UNVERIFIED) Корпорация Майкрософт
Version: 5.1.2600.5512
Image file name:
C:\Users\hrp\AppData\Local\Temp\8D6D.tmp

Process

Command line: "C:\Users\hrp\Downloads\xpajB.exe"

Current directory: C:\Users\hrp\Downloads\

Started: 6 minutes and 13 seconds ago (17:29:48 16-10-2025)

PEB address: 0x3f6000 (32-bit: 0x3f7000) Image type: 32-bit

Parent: explorer.exe (3688)

Mitigation policies: None

Protection: None

Permissions Terminate

Close

Observations:

Registry Modifications

While the Process Monitor is filtered to show **File System activity**, the overall behavior of **xpajB.exe** strongly implies registry interaction, even if not directly visible:

- **Implied Registry Probing:** The process exhibits intense system-wide probing for files (NAME NOT FOUND for DLLs), which is often preceded or accompanied by similar probing in the **Registry** (e.g., searching for application paths, configuration values, or CLSIDs in HKEY_CLASSES_ROOT or HKEY_LOCAL_MACHINE\Software).
- **Need for Further Monitoring:** To confirm registry modifications, **Process Monitor** would need to be filtered for **Registry Operations** (e.g., RegOpenKey, RegSetValue, RegCreateKey). Tools like **Regshot** are ideal for a snapshot comparison to capture all modifications made during execution.

Identified Persistence Mechanisms

The analysis of the process properties reveals several key indicators related to execution and persistence:

A. Execution Origin and Persistence

- **Command Line:** The process is launched from a user's download directory:
 - "C:\Users\hrp\Downloads\xpajB.exe"
 - **Implication:** This confirms the file was manually or programmatically placed in the **Downloads** folder, a common staging location. However, this line alone does not show *how* it was launched (e.g., manually, or by a script/loader).

B. Process Properties and Integrity

- **Multiple Instances:** Two instances (PID 4056 and PID 5572) are running simultaneously and are active (CPU usage of **11.43%** and **2.27%** respectively, from Screenshot 173424.png).
 - **Implication:** This suggests the application is either multithreaded (with multiple processes running the same image) or using a **redundant execution method** for reliability.
- **Parent Process:** The parent process for both instances is **explorer.exe (PID 3688)**.
 - **Persistence Mechanism Identified:** Launching a process directly from explorer.exe (the Windows shell) is typical of **user-launched applications** but is also a common mechanism for persistence established via **Run keys** (Registry), **Shell startup folders**, or **Scheduled Tasks** configured to run at logon.

- **Image Type and Description:** The file is a **32-bit** application. The file description is Russian and translates to "**Main application access library**" or similar, and it is marked **(UNVERIFIED) Microsoft Corporation**.
 - **Implication:** This is a key flag for "**Living off the Land**" (LOTL) or **Masquerading**. By using a fake, misleading description and pretending to be an "UNVERIFIED" Microsoft file, the process attempts to blend in with legitimate system components.

C. DLL Hijacking

- **Evidence:** The Process Monitor logs show massive, rapid probing for hundreds of DLLs (NAME NOT FOUND) in system paths (System32, SysWOW64).
- **Implication:** This aggressive DLL search pattern strongly suggests the program is attempting to perform **DLL Search Order Hijacking**—trying to load a malicious component before a legitimate one, or confirming that its necessary components are loaded.

Memory Analysis and Injection Detection

The process exhibits classic behavior suggesting its running memory image is deliberately obfuscated or replaced:

- **Execution from Temporary File Paths:** Both running instances of xpajB.exe are launched from temporary folders:
 - **PID 5572** is running from C:\Users\hrp\AppData\Local\Temp\1229.tmp.
 - **PID 4056** is running from C:\Users\hrp\AppData\Local\Temp\8D6D.tmp.
 - **Implication:** This is common for a **dropper** or **loader** that unpacks the final payload into a temporary location and executes it. The memory of these temporary files is the primary target for analysis, as it will contain the **unpacked, final payload code**.
- **Masquerading/Deceptive Description:** The file description is a deceptive attempt to blend in, claiming to be an "**(UNVERIFIED) Microsoft Corporation**" file with a Russian description that translates to "Main application access library".

Evidence of Memory-Resident DLLs

The aggressive file system probing strongly suggests an attempt to load specific components, which would reside in memory upon success:

- **DLL Sideload/Search:** The Process Monitor logs show the process aggressively searching for numerous DLLs (e.g., ATMLib.DLL, cabinet.dll) across system and user directories (like C:\Users\hrp\Downloads).

Command Execution and Shell Activity

The detailed event logs from Process Monitor were analyzed to detect any command execution or shell activity. The investigation confirmed that the malware and its child processes did not launch any instances of cmd.exe or powershell.exe during the observation period.

Behavioral Indicators Extraction

1. Execution & Process Indicators

Indicator	Source Evidence	Malicious Implication
High CPU Usage	PID 4056 using 11.43% CPU.	High resource consumption indicates intensive processing, which aligns with aggressive file scanning or payload execution.
Temporary File Execution	Running from C:\Users\hrp\AppData\Local\Temp*.tmp (e.g., 1229.tmp, 8D6D.tmp).	Classic behavior of a dropper/loader that unpacks and executes its true payload from an isolated, temporary, and less monitored location.
Masquerading/Deception	File description is a deceptive Russian string and is marked "(UNVERIFIED) Microsoft Corporation".	Attempts to bypass scrutiny by claiming to be a legitimate, unverified system component.
Parent Process	Parent is explorer.exe (PID 3688).	Indicates the program was launched via a user-session trigger, strongly suggesting a Run key or Startup folder persistence mechanism.

2. Registry & Persistence Indicators

Indicator	Source Evidence	Malicious Implication
Registry Modification	63 total changes, including 1 Key added, 3 Values added, and 58 Values modified.	Confirms the process actively manipulates the system configuration database. The value additions/modifications are typically used to set up persistence (Run Keys) or store C2/configuration data.
Persistence Mechanism	Inferred via parent process being explorer.exe.	Strong indicator that the process established persistence through a Run key or Shell Startup item to ensure execution upon user login.

3. File System Indicators

Indicator	Source Evidence	Malicious Implication
Aggressive DLL Probing	Massive volume of CreateFile operations resulting in NAME NOT FOUND for hundreds of DLLs in C:\Windows\System32 and C:\Windows\SysWOW64.	Clear signature of DLL Search Order Hijacking, attempting to find a vulnerable system DLL to load a malicious component.
Targeting Dropped Payloads	Explicit access attempts on files like cabine.dll in the C:\Users\hrp\Downloads folder.	The process is accessing components staged in a user-accessible location, confirming the use of dropped payloads.
Access Denied	CreateFile operation by PID 4056 resulted in ACCESS DENIED (Desired Access: Write).	Indicates a failed attempt to write, modify, or create a file in a protected system or application directory, likely for persistence or configuration.

Conclusion

1. Masqueraded Launch and Persistence: The process is a self-extracting payload running from a user's temporary folder (.tmp files) and is falsely labeled as an "UNVERIFIED Microsoft Corporation" library. It establishes persistence via a user-logon mechanism (launched by explorer.exe) and confirms system changes with 63 registry modifications.
2. Evasion and Sideloaded Prep: It attempts to execute an advanced evasion technique by aggressively probing for hundreds of non-existent system DLLs. This behavior is the signature of preparing a DLL Search Order Hijacking attack to load a malicious component.
3. Malicious Activity: The process actively consumes high CPU resources (11.43%) and attempts to make changes in protected system directories, as evidenced by a blocked ACCESS DENIED write operation.
4. Overall Assessment: xpajB.exe is confirmed to be malware that ensures system longevity (persistence), actively evades security by using temporary files, and prepares for stealthy execution of its primary function by attempting to perform DLL hijacking.